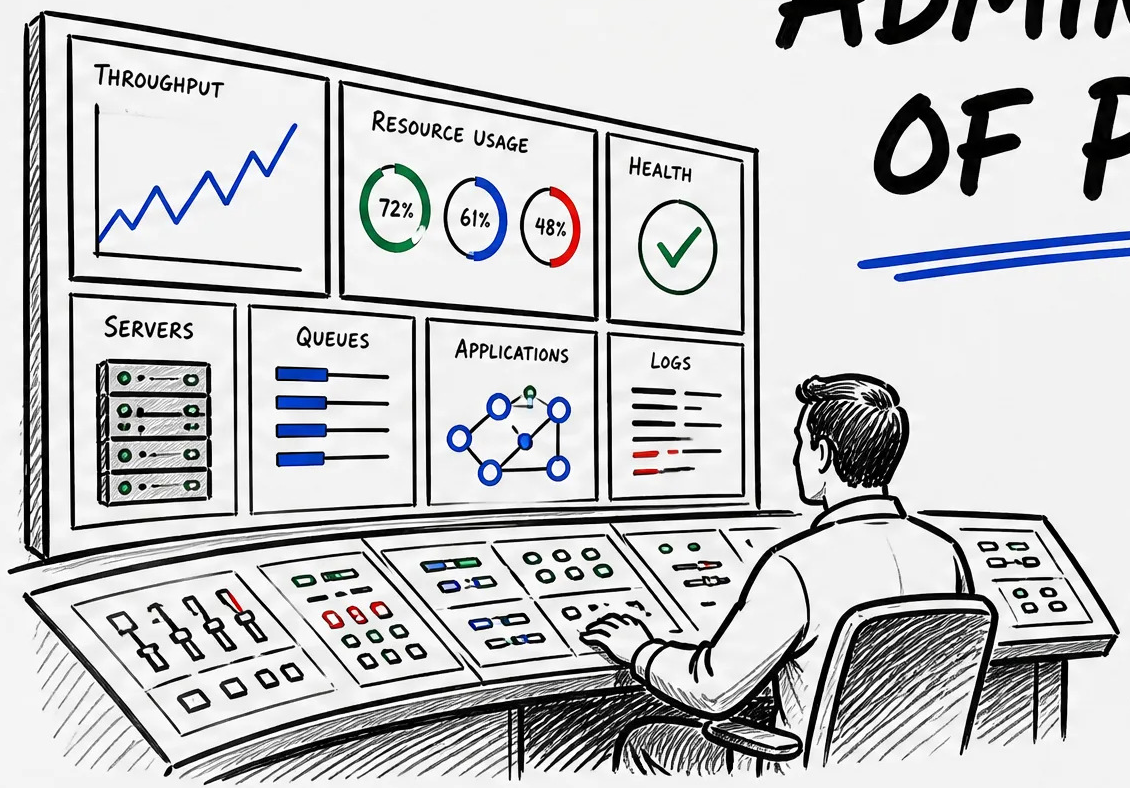
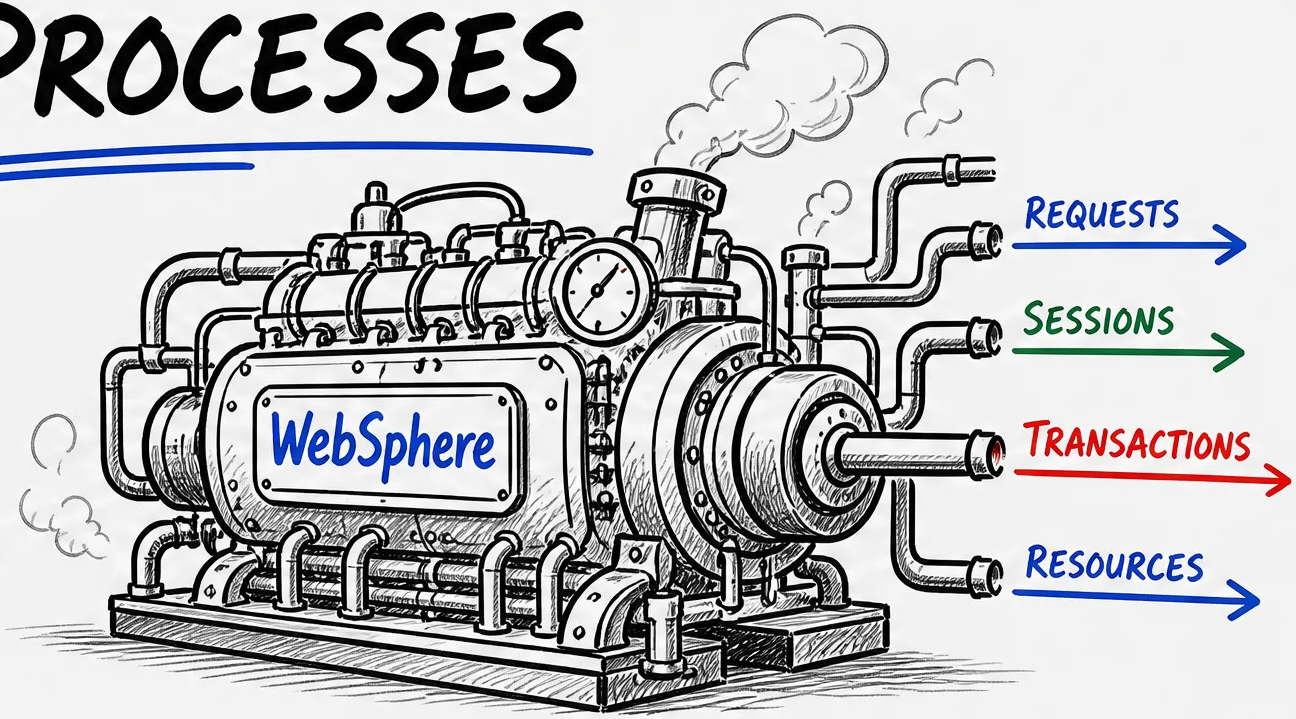
ProTech
protechtraining.com

WebSphere v9 Administration: Module 4

Dr. Lee

ADMINISTRATION OF PROCESSES



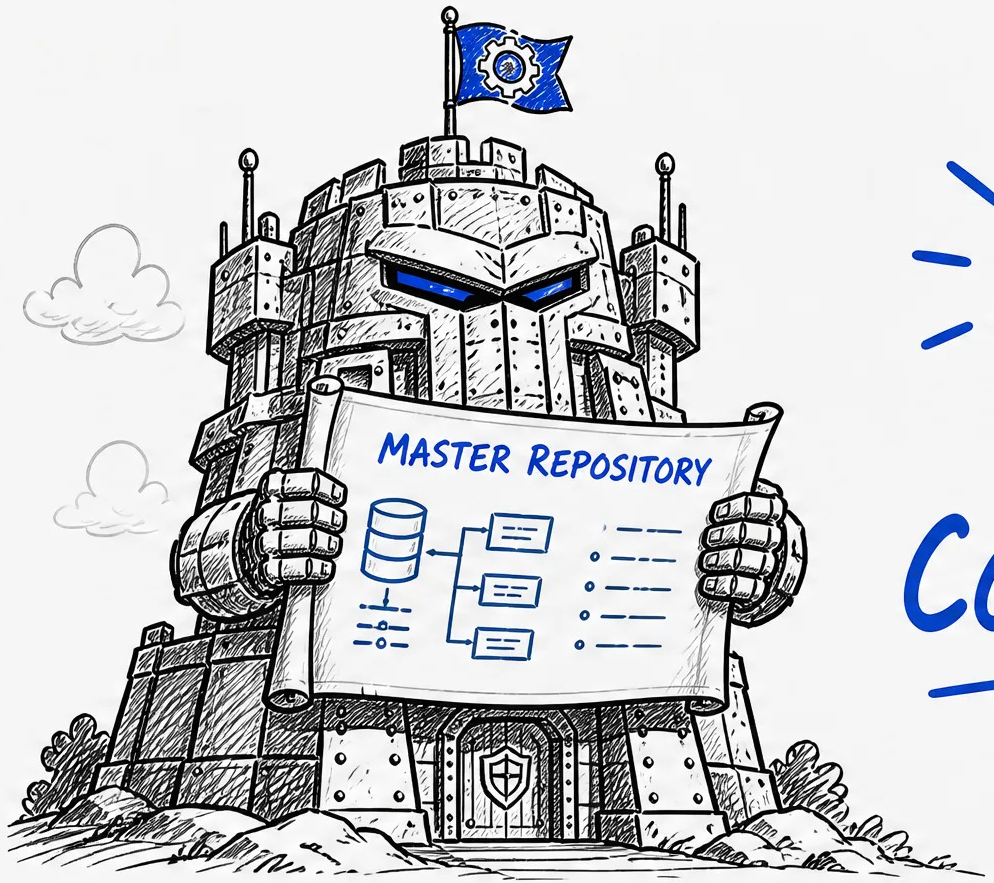


MEMORABLE STATEMENT: Administration is the discipline of continuous control.

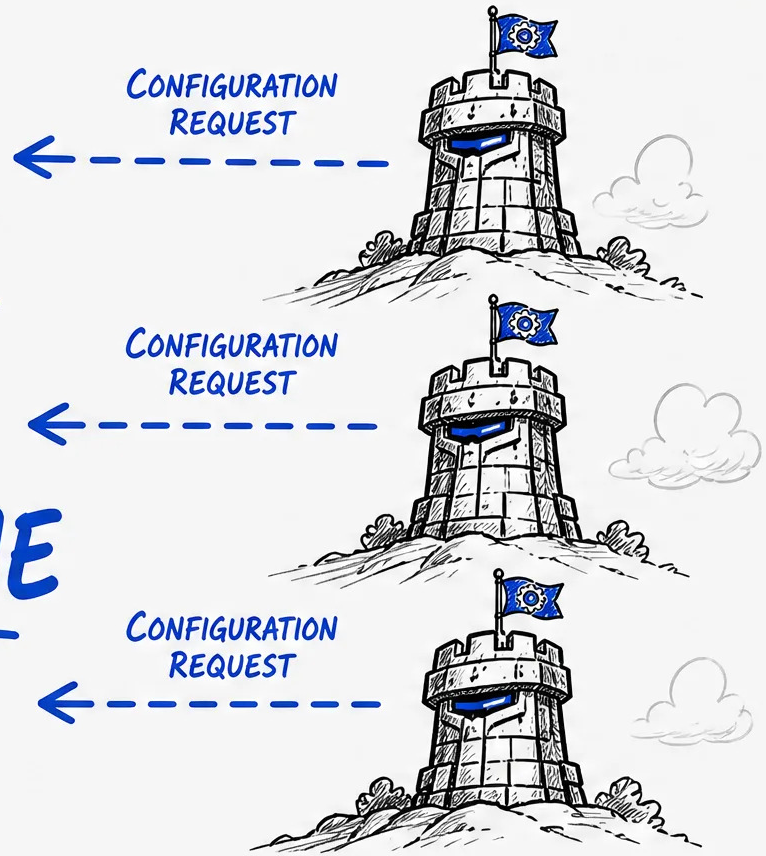
ProTech
protechtraining.com

Deployment Manager Configuration

Dr. Lee



Optimize the Control Plane



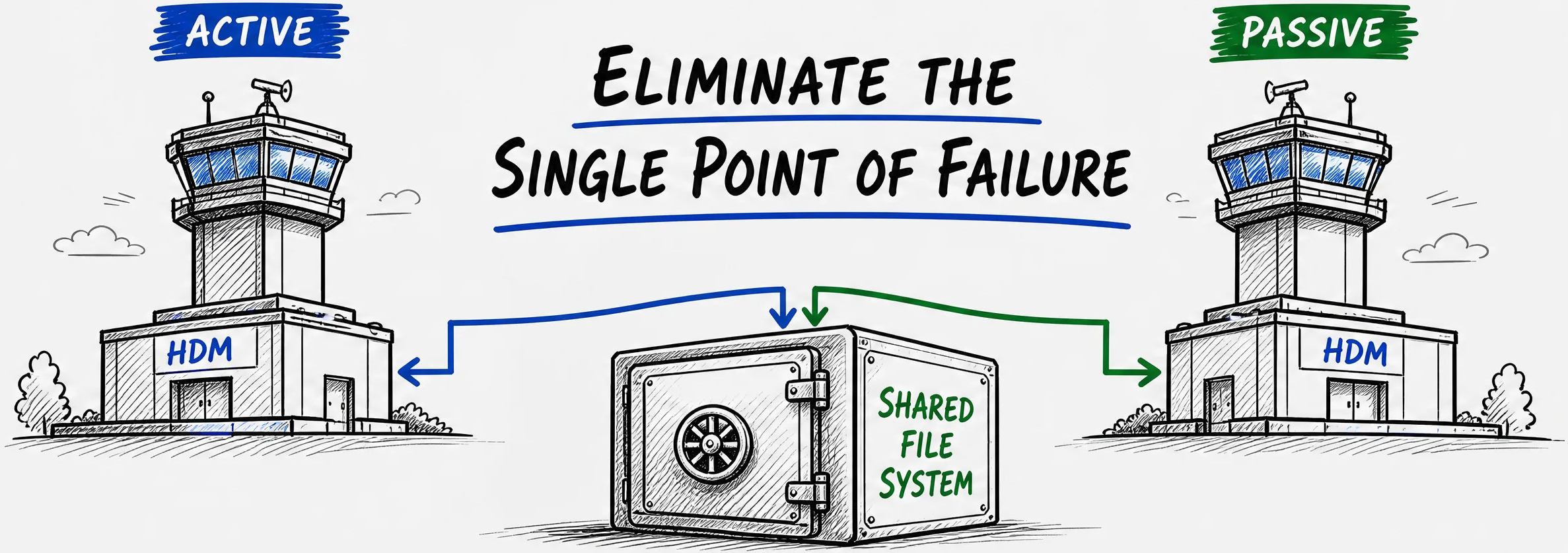
Memorable Statement: The manager must be tuned for stability, not throughput.

ProTech
protechtraining.com

HA Deployment Manager

Dr. Lee

Eliminate the Single Point of Failure



Memorable Statement: HA protects the control plane, not the data plane.

ProTech
protechtraining.com

STARTING AND STOPPING THE DMGR



ADMIN TOOL / SCRIPT
startManager
(Windows Command)

IBM MQ
Windows Service

z/OS COMMAND
START QMGR(MYQMGR)
(TSO/E)

KNOW YOUR HOST ENVIRONMENT

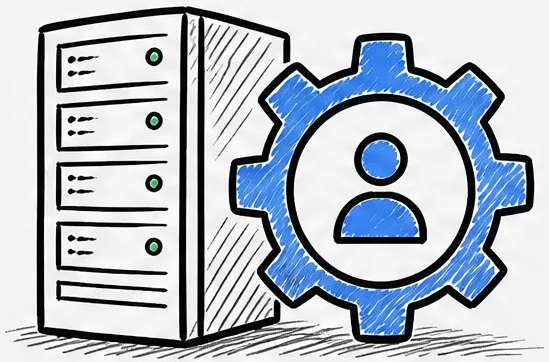
MEMORABLE STATEMENT: Know the native mechanism of your operating system.

Admin Agent and Job Manager Lifecycle

`>_ startServer adminagent`

↓

STARTS PROCESS



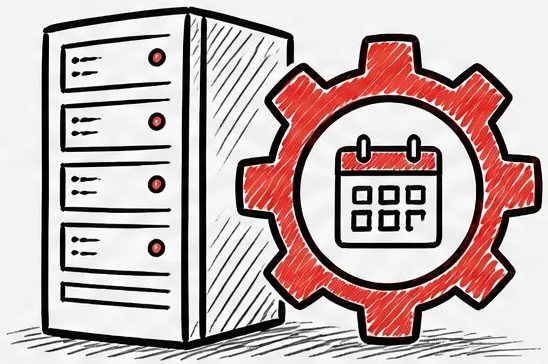
ADMIN AGENT
Server Process

Standard Commands for Flexible Management

`>_ startServer jobmgr`

↓

STARTS PROCESS



JOB MANAGER
Server Process

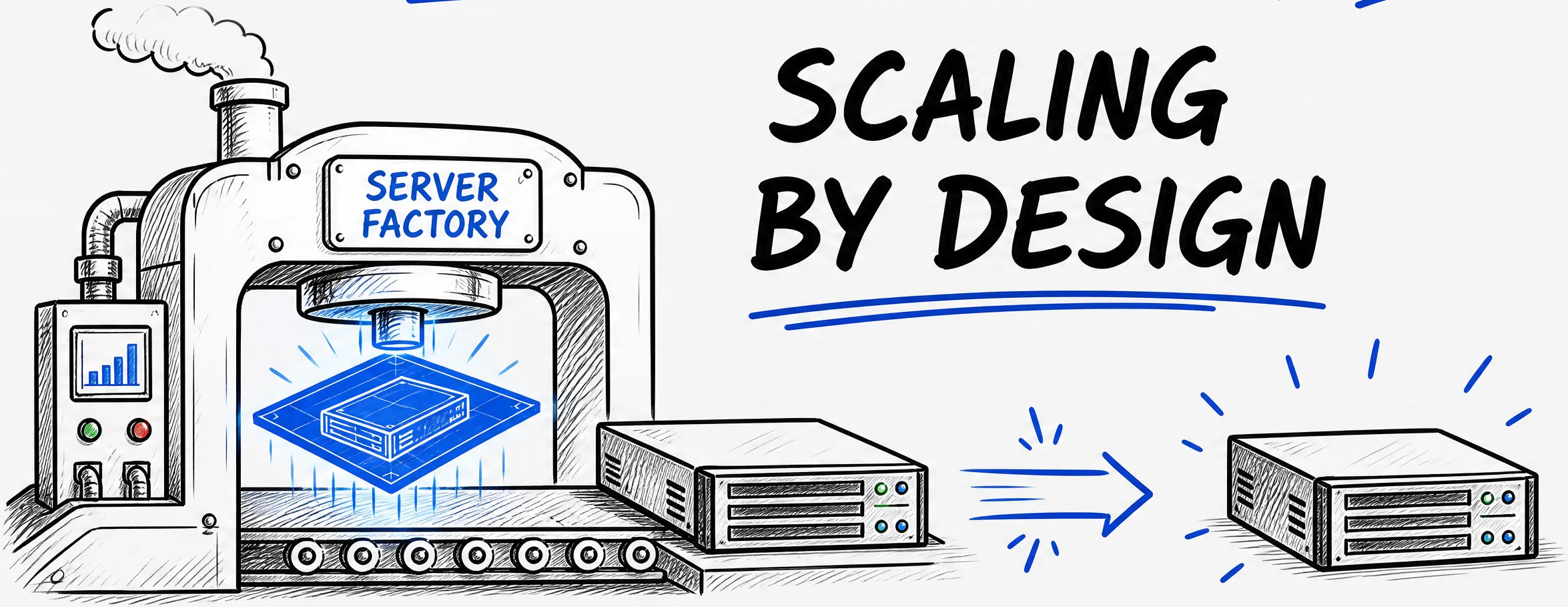
MEMORABLE STATEMENT: Flexible management processes use the standard server commands.

ProTech
protechtraining.com

Application Server Creation

Dr. Lee

SCALING BY DESIGN



☆ **MEMORABLE STATEMENT:** Templates ensure consistency across the topology.

ProTech
protechtraining.com

Starting and Stopping Servers

Dr. Lee

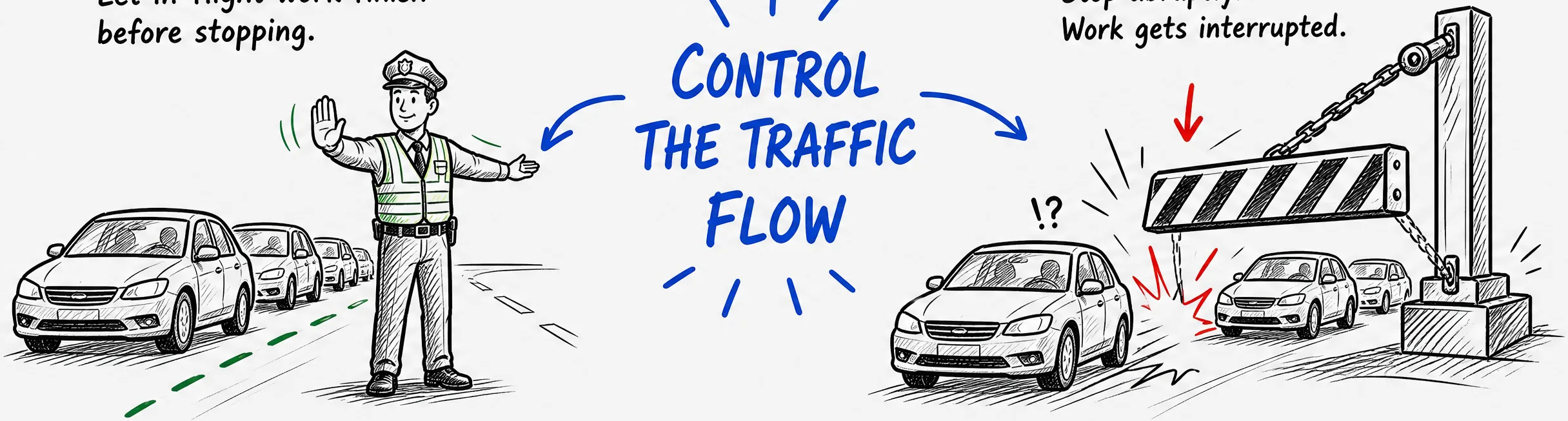
Graceful Stop

Let in-flight work finish before stopping.

Immediate Stop

Stop abruptly.
Work gets interrupted.

Control the Traffic Flow

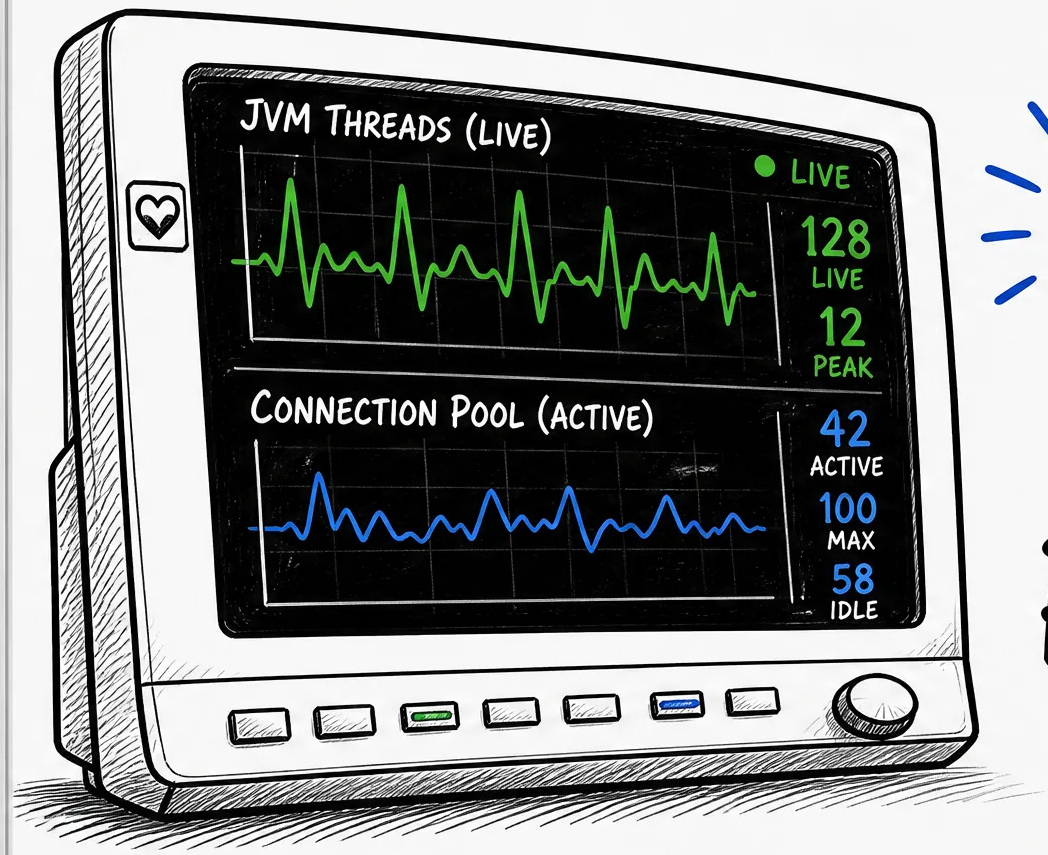


Memorable Statement: A graceful stop protects user transactions.

ProTech
protechtraining.com

Server Runtime Attributes

Dr. Lee



The Pulse of the Process

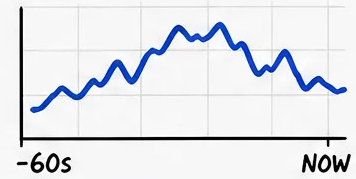
JVM Threads

- Live: 128
- Peak: 12
- Daemon: 98



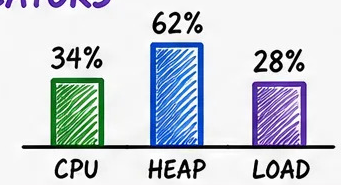
Connection Pool

- Active: 42
- Idle: 58
- Max: 100



Health Indicators

- ✓ CPU: 34%
- ✓ Heap: 62%
- ✓ System OK

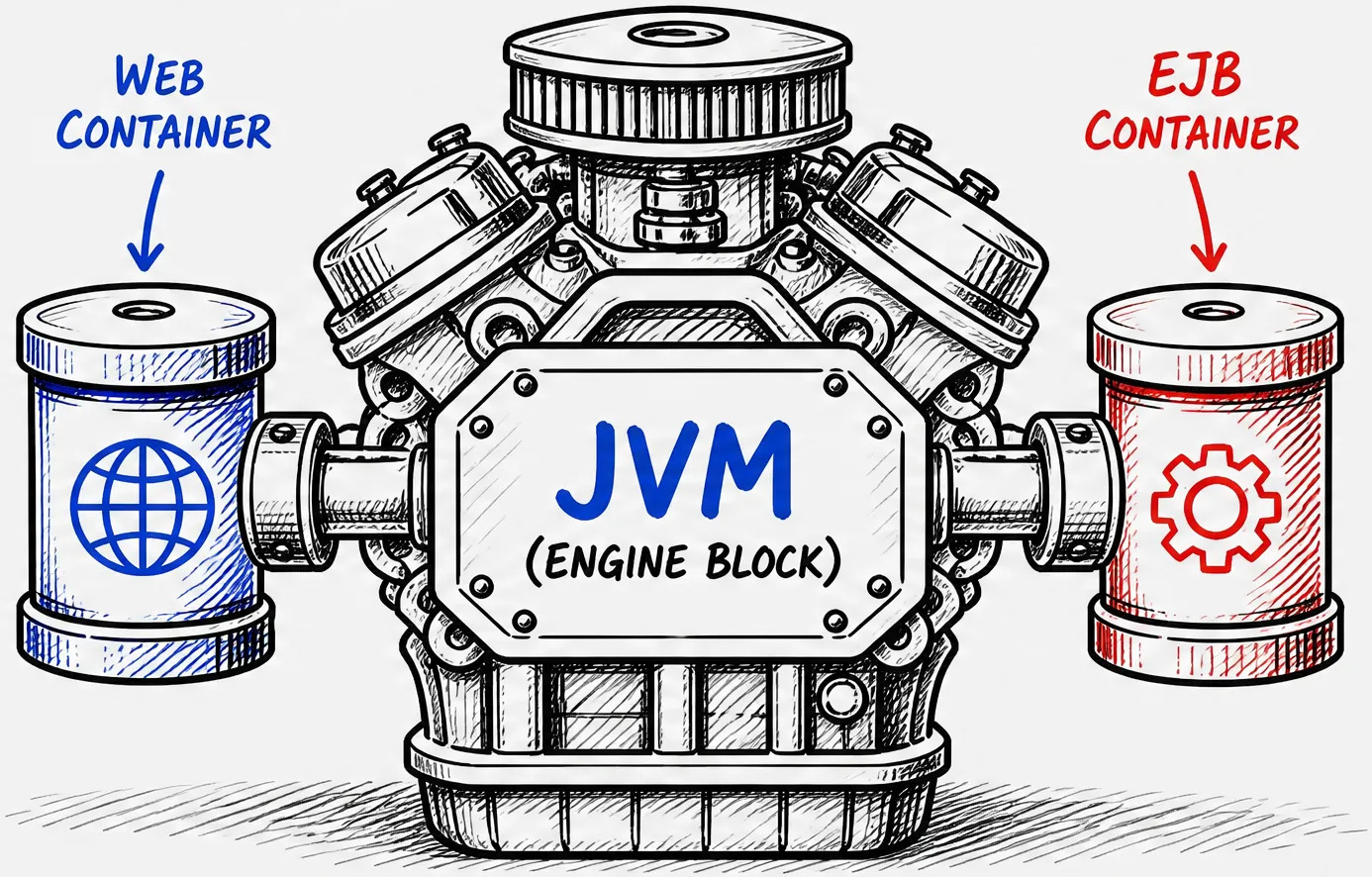


Memorable Statement: The Runtime tab is your window into the living process.

Customizing Application Servers

Dr. Lee

Tuning the Engine



Memorable Statement:

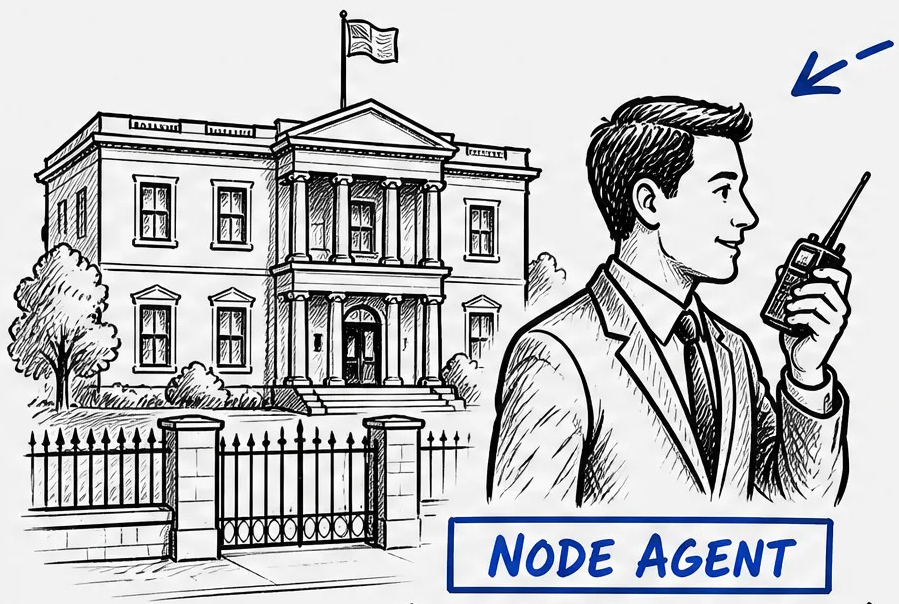
The JVM is the engine block; the containers are the cylinders.

ProTech
protechtraining.com

Node Agent Lifecycle

Dr. Lee

Embassy
(Remote Location)

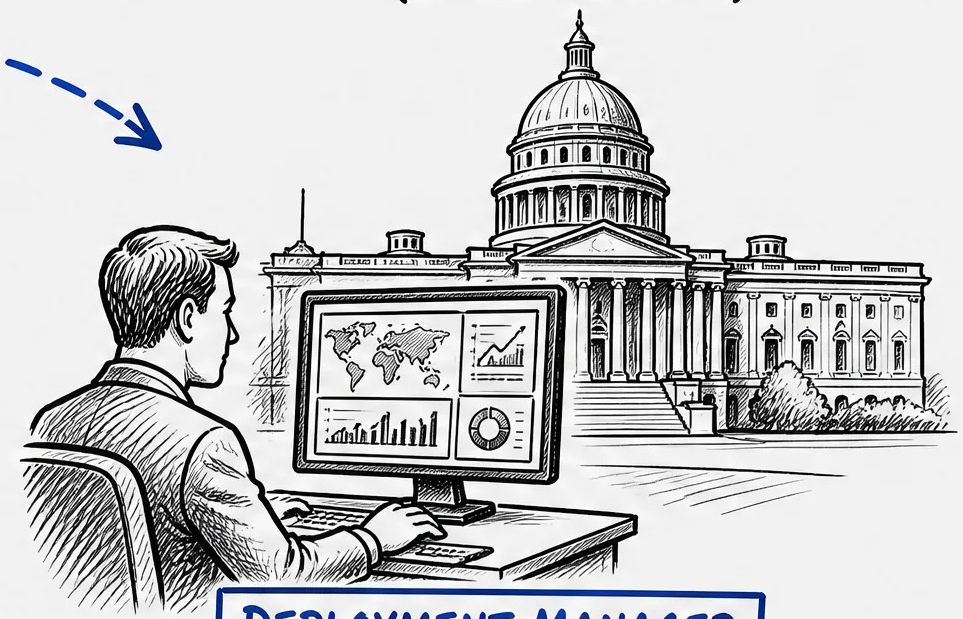


Node Agent
(Embassy Representative)



The Local Authority

Capital
(Central Location)



Deployment Manager
(At the Capital)

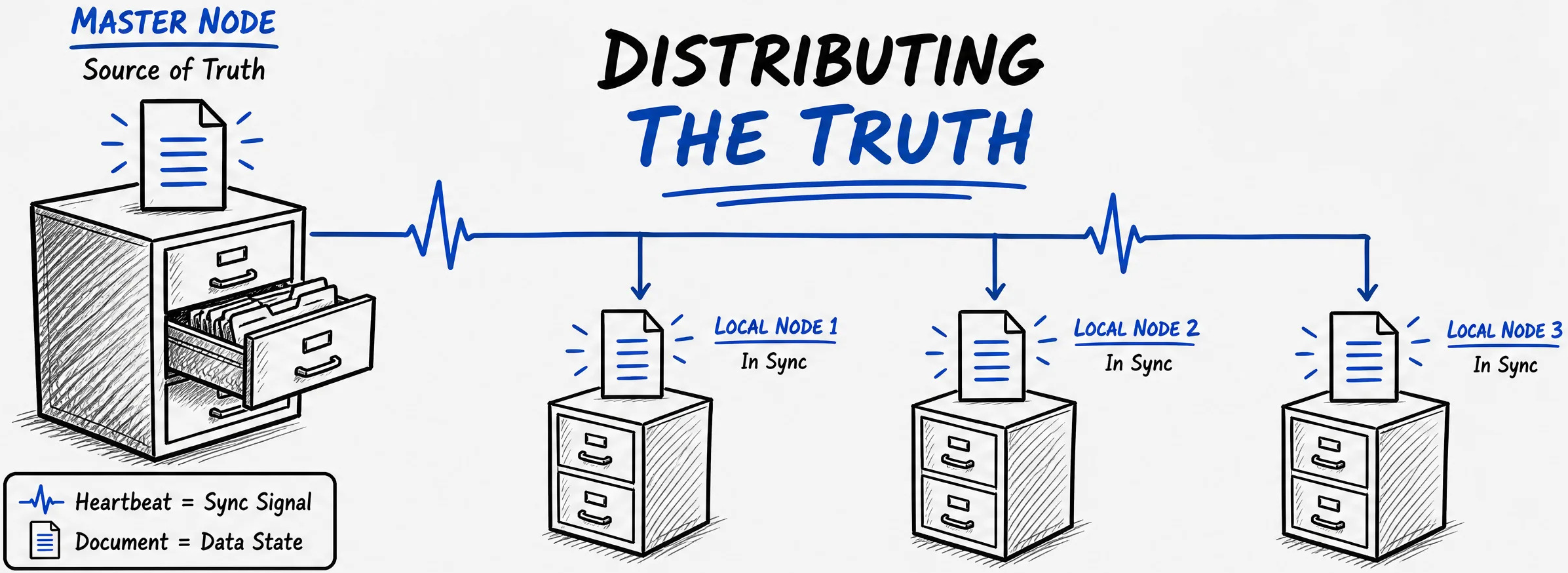
☆ **Memorable Statement:** Without a node agent, remote administration is severed.

ProTech
protechtraining.com

Node Synchronization

Dr. Lee

Distributing the Truth



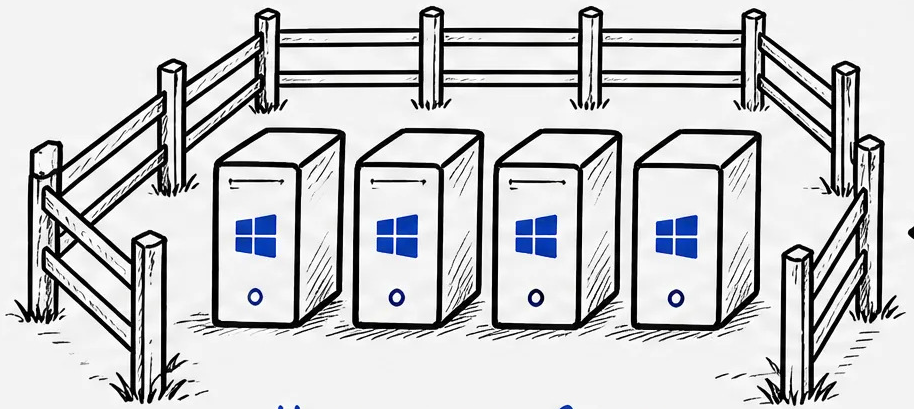
Memorable Statement: Synchronization is the heartbeat of cell consistency.

ProTech
protechtraining.com

Node Groups

Dr. Lee

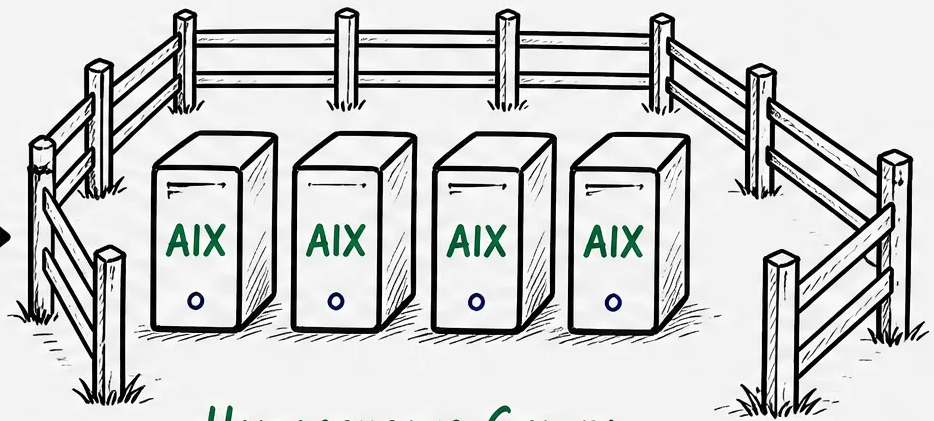
Windows Nodes



Homogeneous Group:
Windows Architecture

Defining the Boundaries

AIX Nodes



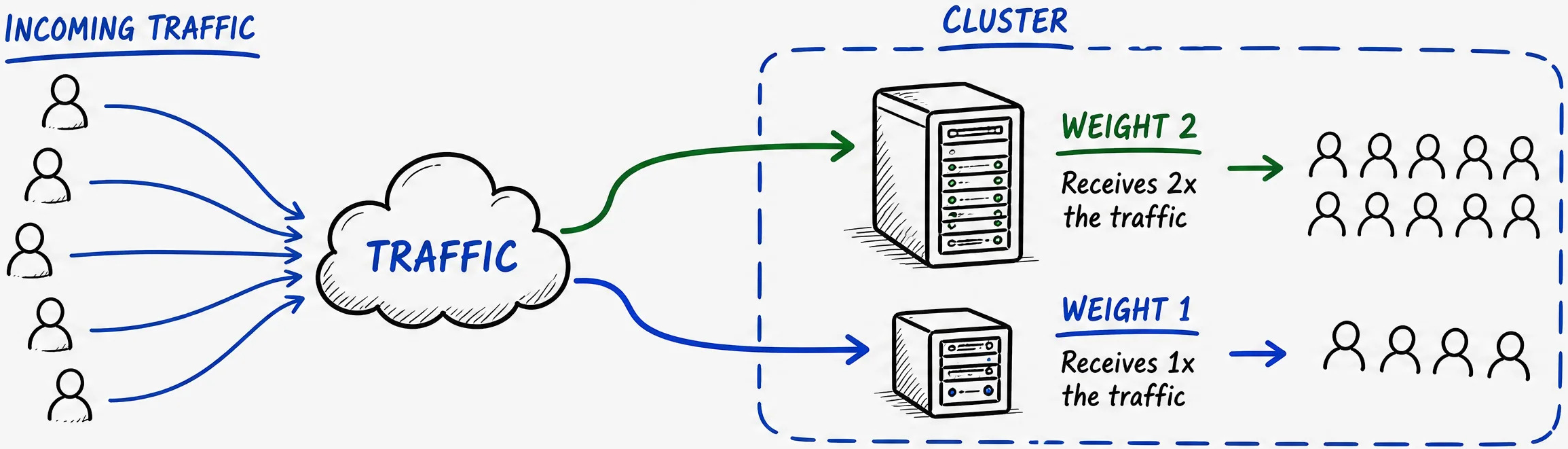
Homogeneous Group:
AIX Architecture

Memorable Statement: Node groups prevent you from clustering incompatible architectures.

ProTech
protechtraining.com

Working with Clusters

Dr. Lee



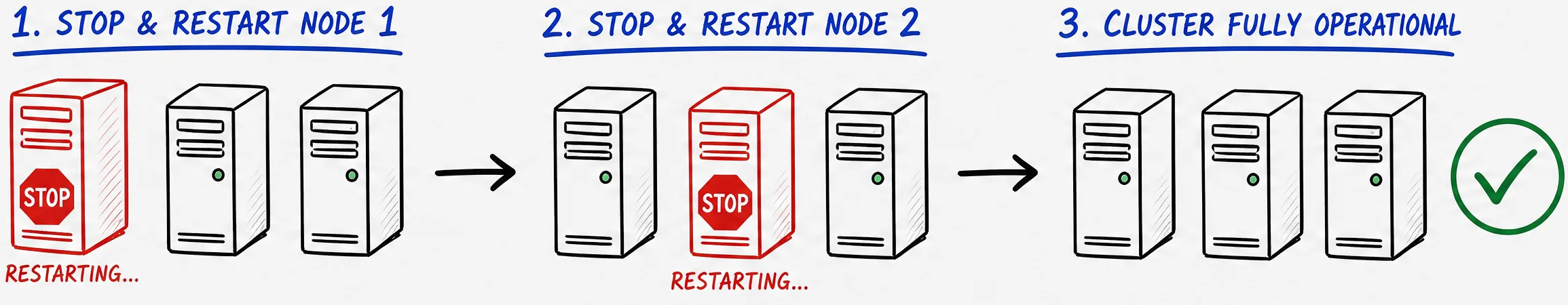
Intelligent Workload Routing

MEMORABLE STATEMENT: Clustering is the foundation of enterprise resilience.

ProTech
protechtraining.com

Managing Cluster State

Dr. Lee



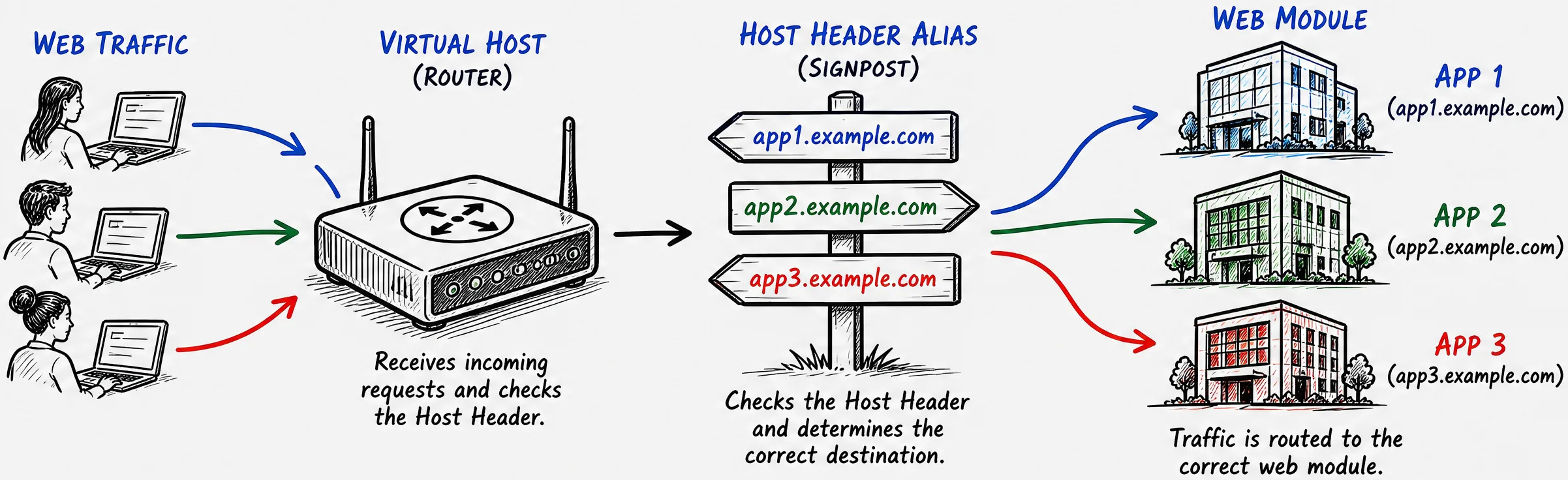
ZERO-DOWNTIME MAINTENANCE

MEMORABLE STATEMENT: Ripplestart is zero-downtime maintenance.

ProTech
protechtraining.com

Virtual Hosts

Dr. Lee

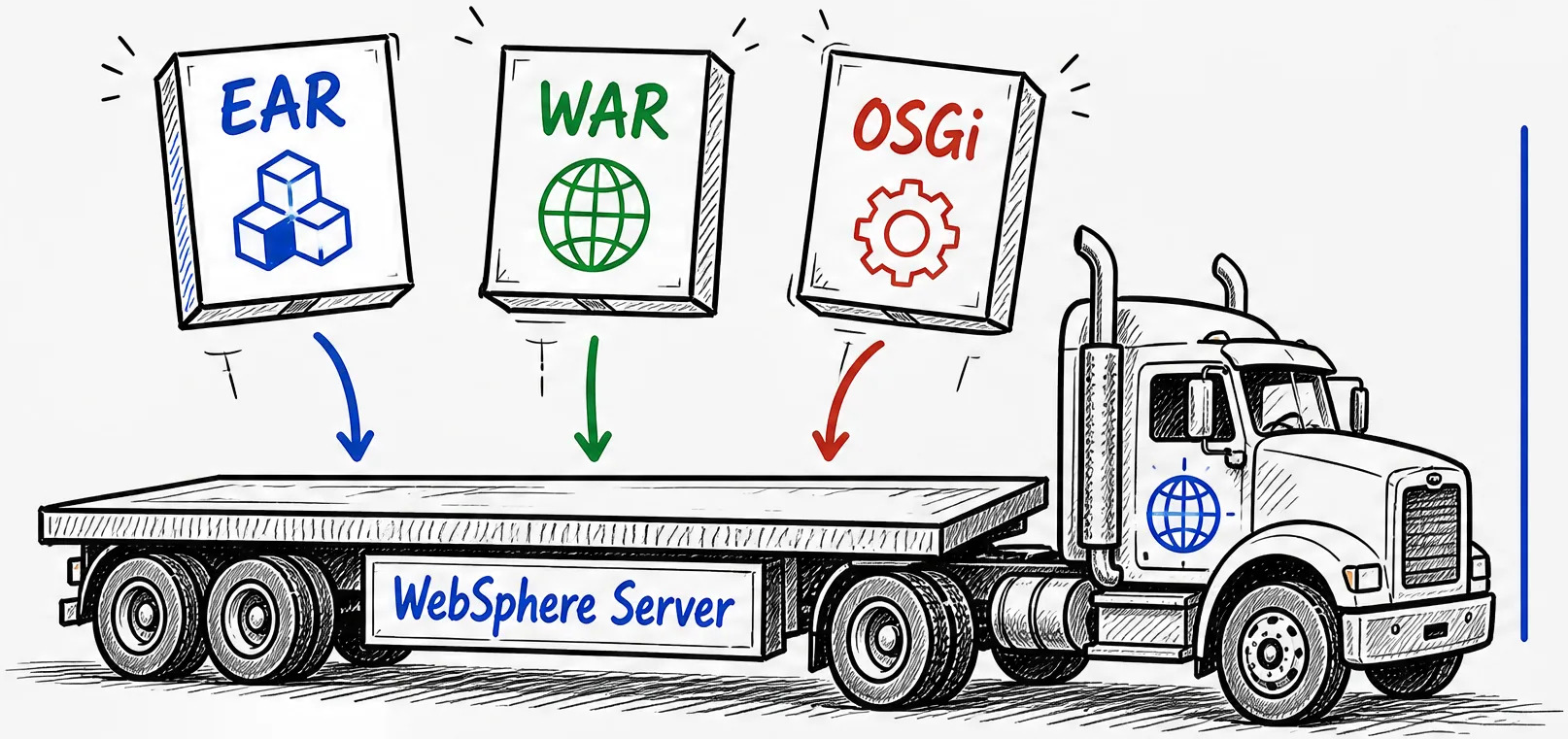


Directing the Network Traffic

Memorable Statement: Virtual hosts route network traffic to the right application.

Managing Applications

Dr. Lee

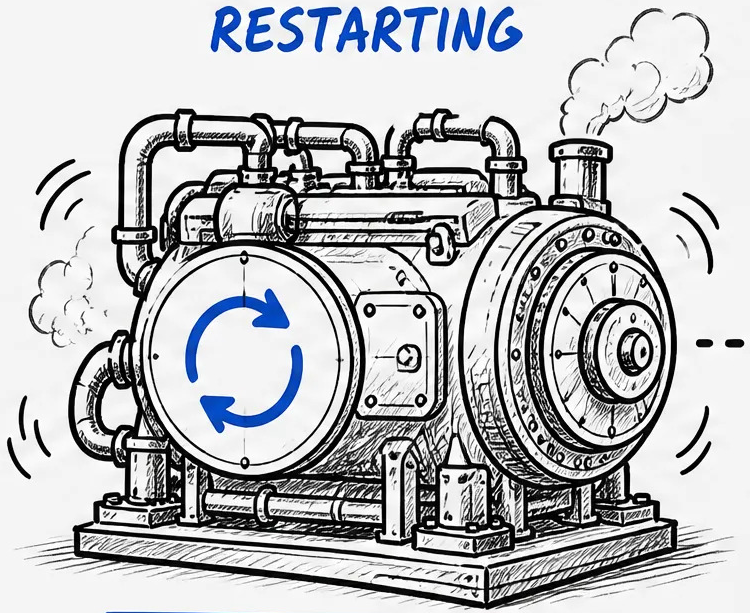


The Payload and the Vehicle

☆ **MEMORABLE STATEMENT:** The application is the payload; WebSphere is merely the delivery vehicle.

Application Lifecycle

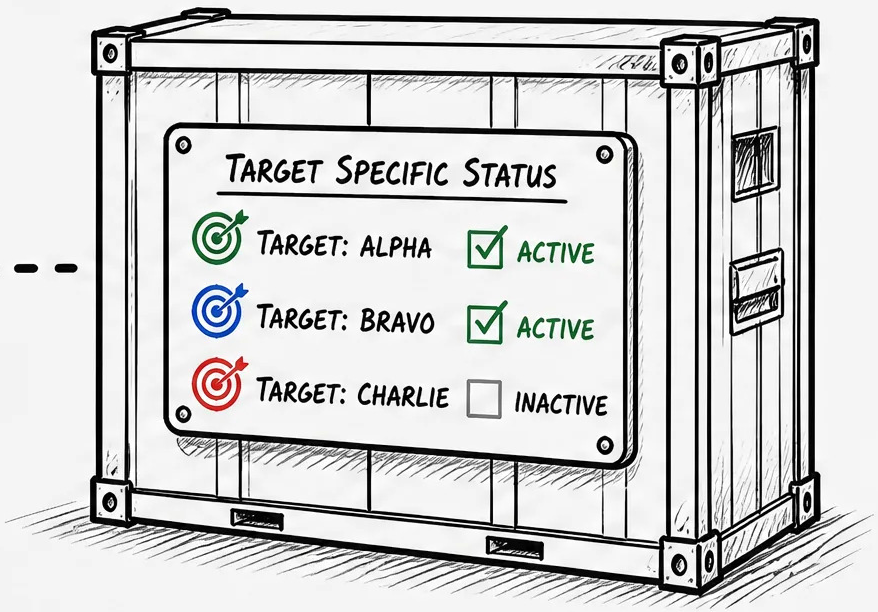
Server Engine Restarting



RESTARTING...

Controlling Payload Ignition

Payload Container



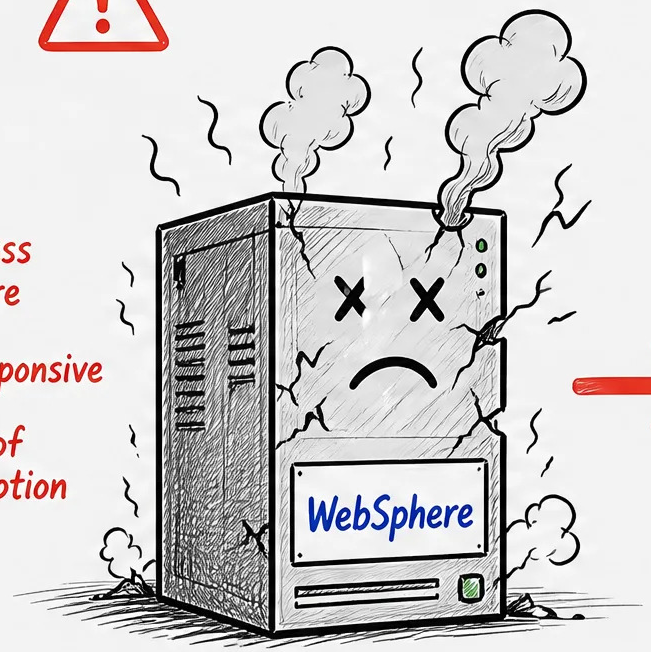
Memorable Statement: You control whether the payload launches with the vehicle.

ProTech
protechtraining.com

Process Restart on Failure

Dr. Lee

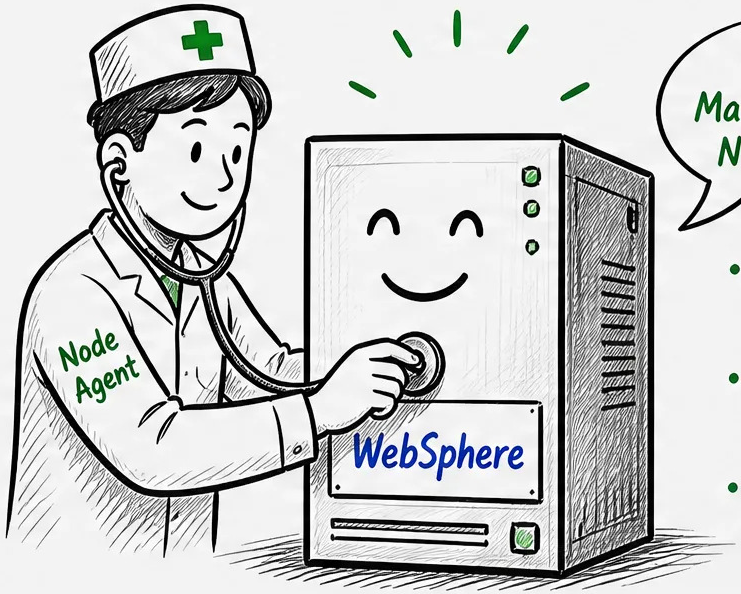
Failure Detected



- Process Failure
- Unresponsive
- Risk of Disruption

Proactive Healing

Instant Recovery & Process Restarted!



- Process Restarted
- Service Restored
- Users Stay Unaware

Memorable Statement: Proactive health management heals the system before the user notices.

ProTech
protechtraining.com

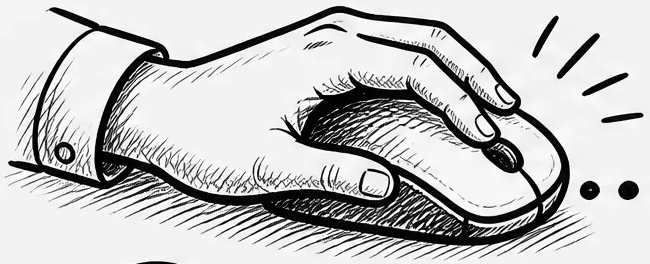
Scripting with wsadmin

Dr. Lee

AUTOMATION AT SCALE

MANUAL CLICKS

SLOW • REPETITIVE • ERROR-PRONE



ONE CLICK AT A TIME

WSADMIN / JYTHON

FAST • CONSISTENT • RELIABLE



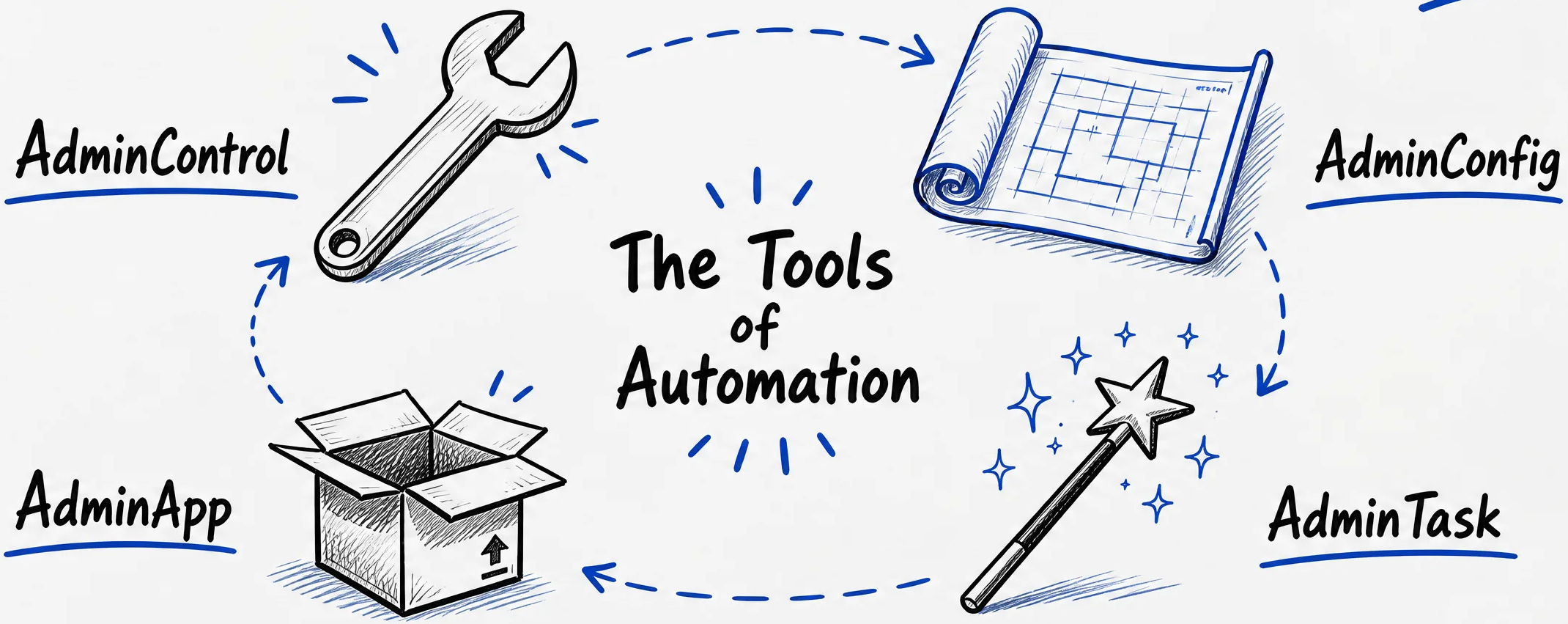
- AUTOMATE TASKS
- REDUCE HUMAN ERROR
- SCALE WITH CONFIDENCE

MEMORABLE STATEMENT: If you have to click it more than twice, you should script it.

ProTech
protechtraining.com

The wsadmin Objects

Dr. Lee



Memorable Statement: Master the objects to master the cell.

ProTech
protechtraining.com

Local vs. Remote Scripting

Dr. Lee

REMOTE (Routine)

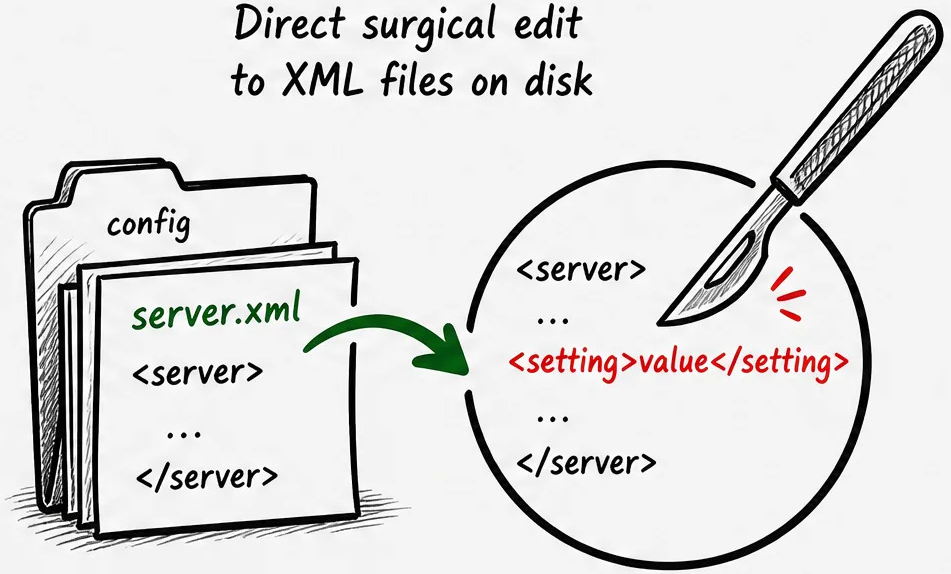
Connect securely via SOAP
to a running server



Rescue
vs.
Routine

LOCAL (Rescue)

Direct surgical edit
to XML files on disk



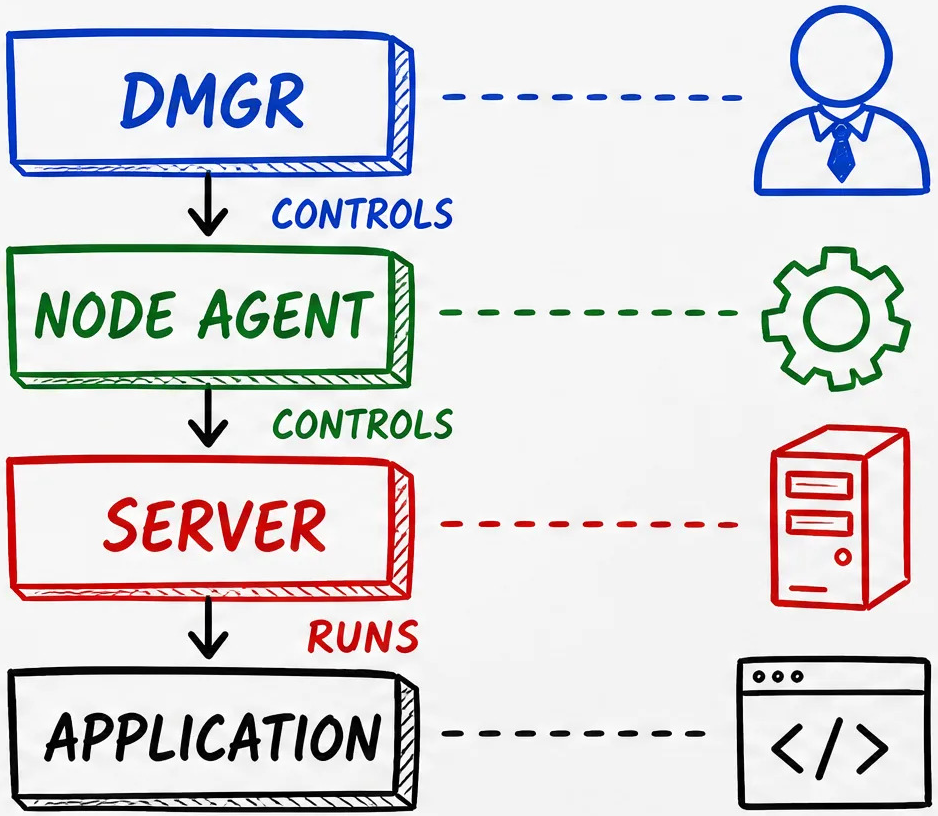
✓ Direct Control ✓ No API Needed ✓ Use with Caution

☆ MEMORABLE STATEMENT: Use local mode for rescue, and remote mode for routine.

ProTech
protechtraining.com

Module 4 Summary

Dr. Lee



KNOW THE HIERARCHY

MEMORABLE STATEMENT: If a lower level fails, check the level immediately above it.